

Greening Namibia's energy future: the case for decentralised renewable electrification

Essay 4 — Greening the energy future

1. Introduction

Ensuring universal access to affordable, reliable and clean electricity is central to Namibia's development strategy. The country's long-term plans, from Vision 2030 to the Sixth National Development Plan, link energy access to poverty reduction, industrialisation, rural development and environmental sustainability (National Planning Commission, 2025; Vision 2030, 2016). Yet Namibia also presents a difficult geography for conventional grid electrification. It has world-class solar and wind resources, but one of the lowest population densities in the world; roughly half of its population still lives in rural areas, while the national grid remains concentrated around urban centres, mining areas and a limited number of transmission corridors (Namibia Statistics Agency, 2024; IEA, 2024).

This essay examines how Namibia can reach universal clean electricity access within a ten-year horizon, while meeting not only household needs but also the energy requirements of productive sectors. It argues that decentralised energy systems should play a central role in this strategy. Grid expansion will remain necessary for towns, mines and large industrial loads, but it is poorly suited to reaching every remote settlement quickly and at reasonable cost. Solar home systems, solar-battery mini-grids and productive-use energy hubs are better matched to Namibia's dispersed rural population, strong solar resource and development priorities.

2. Development context and the access gap

Namibia is an upper-middle-income country of just over three million people, but national income figures conceal significant development constraints. The 2023 Census recorded a population of about 3.02 million, almost evenly split between urban and rural areas: around 1.51 million urban residents and 1.51 million rural residents (Namibia Statistics Agency, 2024). This matters for electrification because the rural half of the country is spread across a very large land area, often far from the existing grid.

Electricity access has improved, but it remains incomplete. Household electrification rose from around 49% in 2019 to approximately 59.5% by the mid-2020s, leaving a large share of households without a formal electricity connection (Windhoek Observer, 2026). The national average also hides a sharp spatial divide. Around three quarters of urban households have access, compared with fewer than two fifths of rural households (Namibia Statistics Agency, 2024; Afrobarometer, 2024). Reliability is also uneven: Afrobarometer reports that rural

households are much less likely to receive a regular supply, and many already rely on alternative sources, including solar systems (Afrobarometer, 2024).

This access gap is directly linked to Namibia's development objectives. Vision 2030 presents energy as a prerequisite for industrialisation and poverty reduction, while NDP6 places economic transformation, inclusive growth and sustainable development at the centre of the country's planning framework (Vision 2030, 2016; National Planning Commission, 2025). In practical terms, electricity access affects the quality of health services, schooling, water supply, agricultural productivity and small enterprise creation: a rural clinic needs refrigeration and lighting; a school needs power for digital learning; a farming community needs energy for water pumping, milling, storage and refrigeration. In addition, in a country where youth unemployment remains high and rural poverty is persistent, the value of electrification lies as much in household appliances as in the productive uses that can support local incomes (Namibia Statistics Agency, 2024).

Namibia's institutional context strengthens the case for a coordinated decentralised access strategy. The country has a relatively stable democratic system and a tradition of national development planning, which matters because off-grid electrification depends on local consultation, tariff legitimacy and cooperation between national agencies, regional councils and traditional authorities. This is not just a governance detail: democratic institutions have been shown to support renewable energy deployment, while recent African evidence suggests that participatory forms of democracy, like Namibia, can positively shape how energy access interacts with inequality (Chen, Pinar and Stengos, 2021; Adams, Ofori and Gbolonyo, 2025).

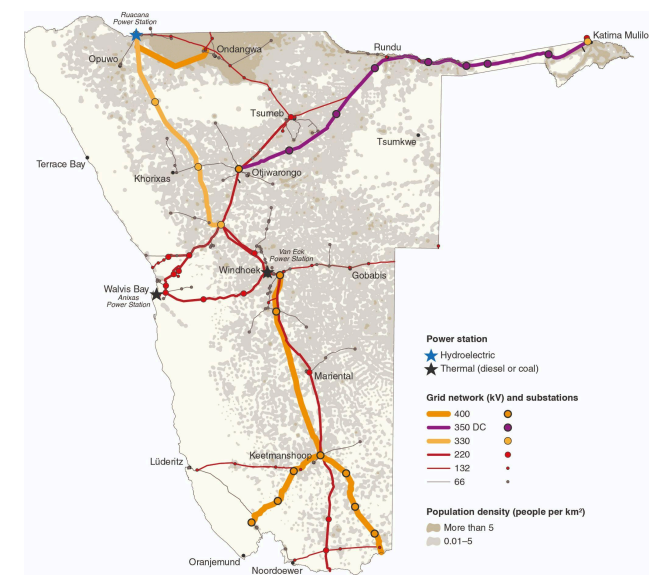
The policy direction is already broadly aligned with clean electrification. Indeed, Namibia's National Energy Compact under Mission 300 sets out a programme to expand access, increase domestic generation and raise the renewable share of electricity supply, with total investment needs estimated at about US\$1.76 billion by 2030 (African Development Bank, 2025; National Energy Compact for Namibia, 2025). The remaining question is now what infrastructure mix can deliver universal access within a decade, and this essay argues that the answer lies in off-grid systems.

3. The spatial mismatch: why grid extension cannot do this on its own

Namibia's electrification challenge is shaped by distance. The country covers more than 825,000 km², but has an average population density of only about 3.7 people per km² (Namibia Statistics Agency, 2024). Population is concentrated in the north-central regions, around Windhoek, and along selected coastal and mining corridors. By contrast, the west, south and east include large areas of desert or semi-arid land with very low settlement density.

Baseline map: population density, existing grid network and solar irradiation

Figure 1: Power stations, transmission lines and main substations Sources: Atlas of Namibia, 2026



The baseline map illustrates this tension. Solar irradiation is strong across most of the country, but population and infrastructure are unevenly distributed. The existing grid follows a limited network of corridors connecting Windhoek, Walvis Bay, Tsumeb, Ruacana and major mining or urban loads. But most rural settlements sit far from these corridors. Extending medium-voltage lines across long distances involves high fixed costs per kilometre. Those costs are reasonable

where many consumers can be connected along the route, but they rise sharply on a per-household basis when the settlement is small or remote. Typical medium-voltage extension costs in sub-Saharan Africa are often estimated in the range of US\$9,000–22,000 per kilometre, before household connection costs are added (International Growth Centre, 2024; Rockefeller Foundation, 2020). For a dense settlement close to the grid, extension can still be the least-cost option but in Namibia’s case, composed of mostly small rural clusters located tens of kilometres away, the economics quickly deteriorate.

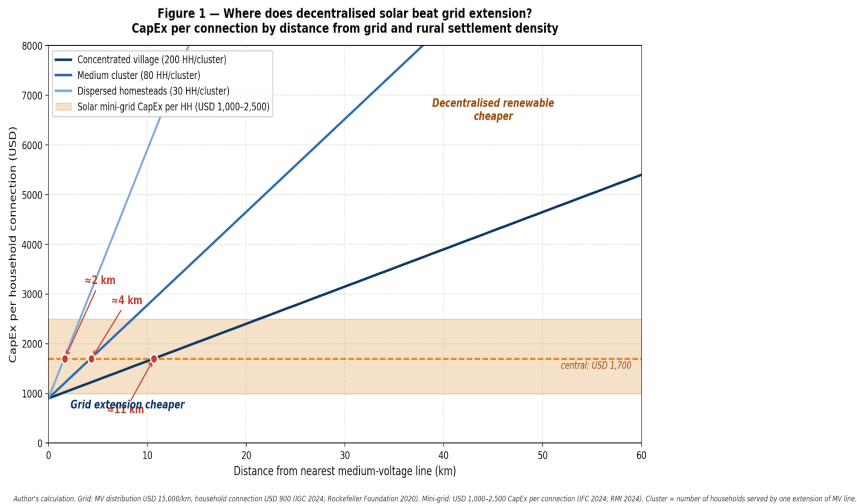
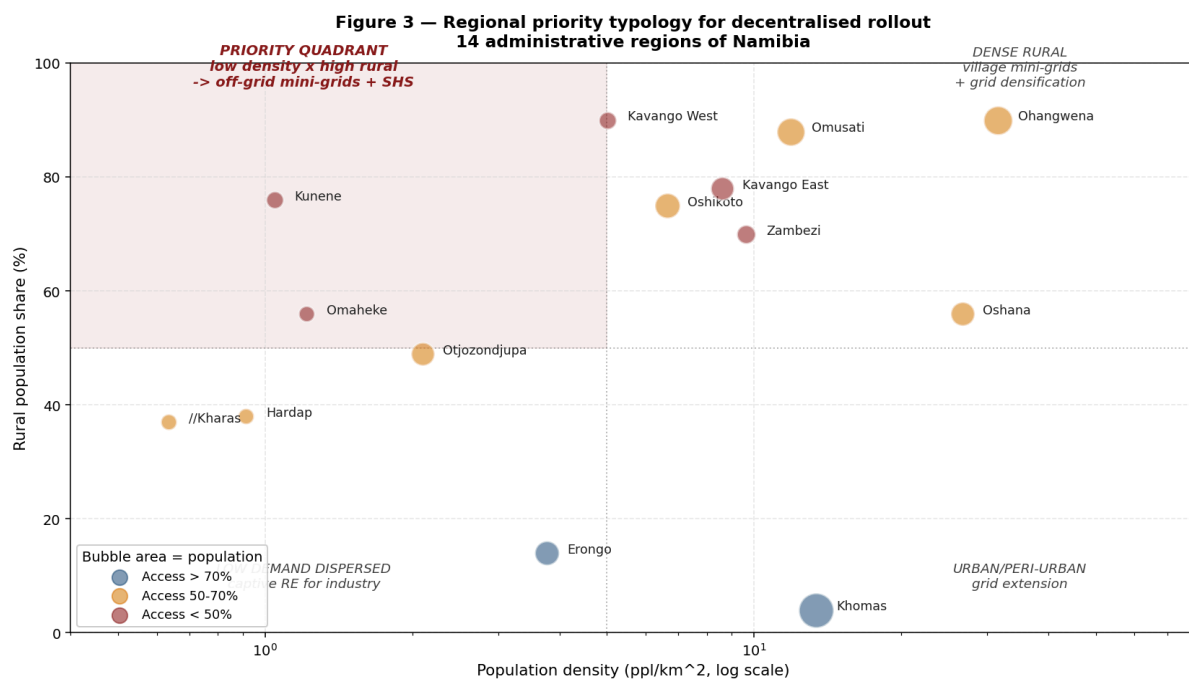


Figure 2: Cost-distance crossover model.

Figure 2 plots the CapEx per household connection of two pathways, extending the medium-voltage grid versus building a solar mini-grid, as a function of distance from the nearest grid line and the size of the settlement being served. Grid costs combine a per-km line cost ($\approx$ USD 15,000, IGC 2024; Rockefeller Foundation 2020) amortised over the cluster, plus a per-household connection cost ($\approx$ USD 900, ESMAP); the mini-grid sits at an all-in USD 1,000–2,500 per connection (IFC 2024; RMI 2024). The result shows that the choice between grid and off-grid infrastructure depends mainly on distance and settlement size. For a village of 200 households, grid extension remains competitive up to roughly 11 km. For clusters of 30–80 households, the crossover falls to between 2 and 4 km. This is particularly relevant in Namibia, where many unserved communities are small, rural and far from existing infrastructure.

The same logic applies at regional level. Ohangwena, Omusati, Oshikoto and Oshana have relatively higher population densities and may justify a combination of grid densification and mini-grids. However, regions such as Kunene, Omaheke, Kavango West and Zambezi have lower densities or more dispersed settlements, but yet have strong rural population share, making long-line grid extension less attractive. In these areas, decentralised systems can reach households faster and with less exposure to the cost and delay of network expansion.



Author's analysis. Sources: NSA 2023 Census Main Report; Afrobarometer Round 9 (2024); AfDB Mission 300 Compact (2025).

Figure 3: Regional priority typology. Population and area: NSA (2024); access categories triangulated from NSA Census 2023 and Afrobarometer Round 9 (2024).

4. Resource base and the structure of future demand

Namibia's renewable resource base is exceptionally strong. Most of the country receives very high solar irradiation, often above 2,700 kWh/m² per year, giving solar PV a much higher productivity than in most European contexts for example (IEA, 2024). Wind resources

are more spatially concentrated, but the south-western coast around Lüderitz has particularly strong potential, with high capacity factors in suitable sites (IEA, 2024). Hydropower plays an important role through the 347 MW Ruacana plant, but its output is vulnerable to rainfall variability and drought, which limits its reliability as the backbone of a universal access strategy (IEA, 2024).

The resource base therefore favours solar as the primary technology for decentralised access, with wind used selectively where the resource is strong. Another benefit of solar PV is that it is modular, quick to install and scalable from household kits to village mini-grids and larger productive-use systems. This makes it well suited to Namibia's geography.

Future demand, however, will not be uniform. Namibia's peak electricity demand is expected to rise substantially, from about 637 MW in 2023 to more than 1,200 MW by 2040 (National Planning Commission, 2025). Part of this growth will come from cities, mining and industrial projects. Mining already represents a large share of electricity consumption, while planned green hydrogen and ammonia projects near Lüderitz could require very large dedicated renewable supplies (NamPower, 2024; Bloomberg, 2025). These loads will require different forms of decentralization such as requiring captive renewables, dedicated grid connections or large-scale procurement.

The rural access agenda involves smaller but developmentally important productive loads. Rural households initially need electricity for lighting, phone charging, refrigeration, communication and appliances. Over time, demand should grow into productive uses: irrigation pumps, cold storage, milling, welding, small shops, clinics, schools, tourism facilities and digital services. This is where solar-battery mini-grids become particularly important. They can provide more than basic household access while also supporting income-generating activities that improve the economics of the system. Daytime productive uses are especially valuable because they absorb electricity when solar output is highest, reducing reliance on batteries and raising utilisation. Namibia therefore needs a segmented approach, in which each technology is matched to the demand it is best able to serve.

5. The case for decentralised clean energy

As explained, the case for decentralised electrification follows from the interaction between Namibia's geography, resource base and development objectives. The rural access gap is dispersed, solar resources are widespread, and the ten-year horizon makes speed essential.

Solar home systems should be used for the most isolated homesteads, especially where households are too scattered to justify a shared network. These systems can provide basic electricity services at low upfront infrastructure cost and can be deployed quickly through pay-as-you-go or subsidised ownership models (IEA, 2024; Falchetta et al., 2023). Their limitation is capacity. They are useful for basic household access, but less appropriate for productive uses that require higher and more reliable loads.

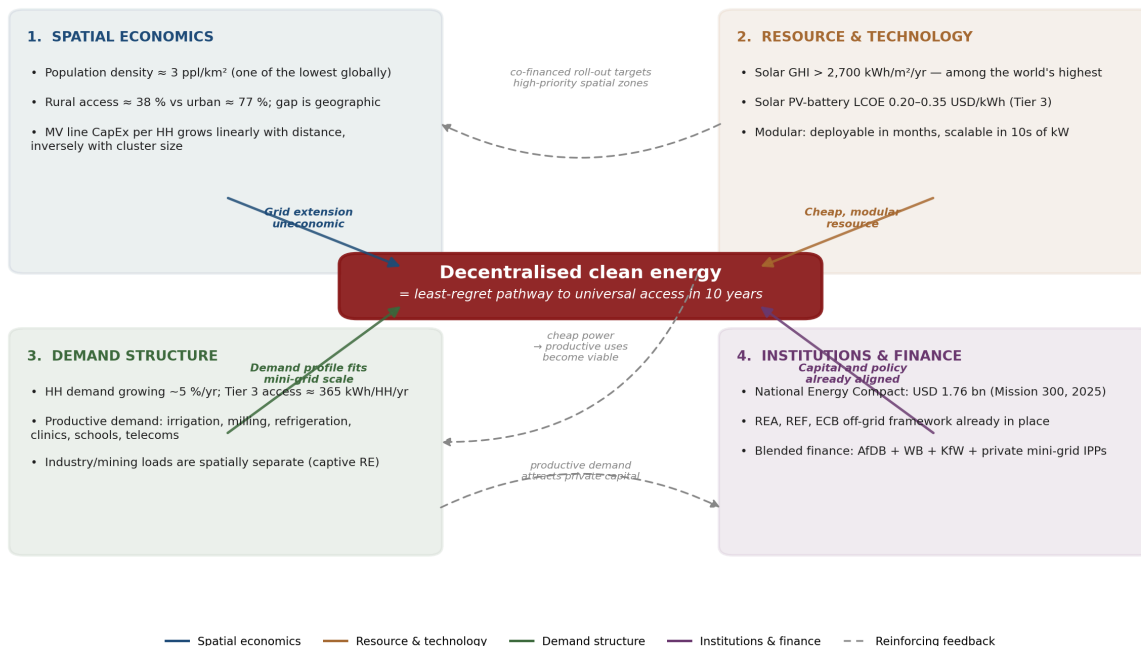
Solar-battery mini-grids should therefore form the centre of the rural access strategy. They are better suited to villages or settlement clusters because they can aggregate demand, support productive loads and allow capacity to expand over time (Falchetta et al., 2023; RMI, 2024). A mini-grid can power households, schools, clinics, water pumps, refrigeration and small businesses within the same local system. Mini-grids are also more compatible with community-level planning: local authorities and traditional leaders can help identify anchor loads, organise users and support payment collection.

Productive-use energy hubs should be planned alongside mini-grids. A village mini-grid becomes more viable when it serves daytime loads such as milling, cold storage, irrigation, workshops, telecom towers or health facilities. These uses raise demand during solar production hours, improve the economics of the system and generate local income. Evidence from mini-grid programmes in other African countries suggests that productive-use anchoring can lower effective electricity costs and improve financial sustainability (Falchetta et al., 2023; REA Nigeria, 2024). In Namibia, this approach fits the broader development agenda because it links electrification to agriculture, water security, rural enterprise and service delivery.

There is also a resilience argument. Namibia imports a large share of its electricity through the Southern African Power Pool, exposing the country to regional supply shortages and price pressures (US Trade.gov, 2024).

The strongest objection concerns affordability. Mini-grid electricity can be more expensive per kWh than grid electricity, especially when demand is low or payment collection is weak. Recent estimates suggest rural mini-grid tariffs in sub-Saharan Africa often need to recover costs in the range of US\$0.20–0.40/kWh unless supported by subsidies or concessional finance (Falchetta et al., 2023; RMI, 2024). This is a real constraint in rural Namibia, where poverty rates are higher and ability to pay is limited. It requires to adapt the financing model. The poorest households cannot be expected to pay the full economic cost of last-mile clean electrification.

Figure 2 — Why decentralised clean energy in Namibia: a systems view
 Four disciplinary lenses (spatial economics, resource & technology, demand structure, institutions & finance) converge on a single conclusion. Dashed grey arrows show reinforcing feedback loops between domains.



Source: author's synthesis. Quadrant data drawn from NSA Census 2023; IEA (2024); IRENA (2024); AfDB Mission 300 (2025); Falchetta et al. (2023).

Figure 4: Why decentralised clean energy fits Namibia's access challenge. Author's synthesis using evidence from NSA (2024), IEA (2024), AfDB (2025), Falchetta et al. (2023) and RMI (2024).

Figure 4 summarizes the argument that dispersed demand raises grid-extension costs; strong solar resources make local generation viable; productive uses improve mini-grid economics; and blended finance addresses affordability.

6. Financing and implementation

The financing challenge is large. Namibia's National Energy Compact estimates total investment needs of about US\$1.76 billion by 2030, with a combination of public funding, concessional finance and private participation (African Development Bank, 2025; National Energy Compact for Namibia, 2025). On top of off-grid expansion, grid reinforcement, generation, storage and distributed systems will all compete for capital.

For decentralised energy, the most realistic model is blended finance. Public and concessional capital should cover the parts of the system that cannot be recovered through tariffs. This includes connection subsidies for poor households, viability-gap funding for low-density mini-grids, public-service loads such as schools and clinics, and the administrative capacity needed to regulate and monitor projects. Development partners, including the World Bank, African Development Bank, Germany, the Global Environment Facility and distributed-renewable investment vehicles, can reduce the cost of capital through concessional loans, grants, guarantees and first-loss capital, making rural projects bankable despite weak household purchasing power and uncertain demand (GEAPP, 2025; Namibian Sun, 2026).

Private capital can then finance and operate systems where demand is sufficient. Mini-grid developers and solar home system companies need predictable concession rules, tariff guidance and clarity over what happens if the national grid later reaches their service area. The Electricity Control Board's draft Off-Grid Electrification Policy is therefore important because it can define licensing, tariff principles and interconnection rules (Electricity Control Board, 2019). Without this regulatory clarity, private investors will price in policy risk or avoid remote areas altogether.

Affordability also requires explicit subsidy design. A cost-reflective tariff may be acceptable for some productive users, but it will be too high for many rural households. Namibia's Rural Electrification Fund can act as a cross-subsidy mechanism, using public resources and contributions from higher-consumption users to reduce connection costs and support lifeline tariffs. Kenya provides a useful precedent: its Rural Electrification Programme Fund has financed rural access through government allocations, donor finance and a levy of up to 5% on electricity sales, using the existing power system as a funding base for underserved areas (World Bank, 2015; NewClimate Institute, 2019).

Implementation also depends on whether Namibia can avoid the classic failure points of rural mini-grids. Namibia's own off-grid experience in places such as Tsumkwe and Gam shows that technical viability is not enough if maintenance, user training and community outreach are weak (Hoeck et al., 2022). The electrification programme should therefore link mini-grid procurement to local technician training, clear tariff communication and nearby repair capacity, rather than treating installation as the end of the project. This is also consistent with Namibia's draft Off-Grid Electrification Policy, which stresses the need for clearer rules around licensing, tariffs and institutional responsibility (Electricity Control Board, 2019).

Three risks are especially important. First, demand may be overestimated, leaving mini-grids underused; this can be reduced through phased deployment, anchor loads and pre-investment surveys of productive uses, since higher daytime demand improves mini-grid economics (ESMAP, 2019; SEforALL, 2024). Second, tariffs may become politically contested if off-grid users pay more for a service they perceive as inferior to the grid; Namibia therefore needs transparent subsidy rules and minimum service standards. Third, poor coordination between grid expansion and mini-grid concessions could strand private assets, so concession contracts should include interconnection, compensation or transfer rules if the grid later arrives (EEP Africa, 2018; ESMAP, 2019).

7. Conclusion

Namibia should place decentralised renewable electrification at the centre of its universal access strategy. The main reason is that Namibia's remaining access gap is concentrated in precisely the places where conventional grid extension performs worst: low-density rural areas, dispersed settlements and communities located far from existing transmission corridors. The evidence points toward a segmented off-grid strategy. Solar home systems

are the most suitable option for isolated homesteads where a shared network would be uneconomic. Solar-battery mini-grids should serve villages and rural clusters because they can combine household access with public-service and productive loads. Productive-use hubs are especially important because they make decentralised electrification part of Namibia's wider development strategy. Finally, larger industrial loads, including mining and green hydrogen, also strengthen the case for decentralised clean supply, although through captive renewables and dedicated procurement rather than village-scale systems.

The main risk is therefore not choosing the wrong technology, but failing to build the institutions around it. Mini-grids will struggle if tariffs are unaffordable, if demand is overestimated, if communities cannot access maintenance, or if private operators face uncertainty when the grid later arrives. Namibia's off-grid strategy must therefore be financed through public and concessional capital, supported by targeted subsidies, and governed through clear concession, tariff and interconnection rules, using case studies such as Kenya. If those conditions are met, decentralised renewable systems offer the most credible route to universal clean access within a decade. They fit Namibia's geography, use its strongest renewable resource, and connect electricity access directly to rural livelihoods and productive development.

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